Ling Wang

Address	School of Mathematical Science, Peking University No. 5 Yiheyuan Road, Haidian District, Beijing, P.R.China 100871
Personal data	Male, born July 18, 1998, Chongqing, China
E-mail	lingwang@stu.pku.edu.cn & lwmath@foxmail.com
Homepage	lwmath.github.io

Research interests

Geometric Analysis & Nonlinear PDEs, especialy Monge-Ampère type equations.

Education

2020.9 – Present	**Peking University** – Beijing, China PhD in pure mathematics Mentors: Professor Bin Zhou.
2020.2 – 2020.6	**Beijing International Center for Mathematical Research** – Beijing, China Visiting student, pure mathematics.
2016.9 – 2020.6	**Beijing Normal University** – Beijing, China BA in mathematics and applied mathematics Mentors: Professor Haigang Li.

Publications

[8] W. K. Du, **L. Wang**, Y. Yang. Flat level sets of Allen-Cahn equation in half-space. Preprint.

[7] **L. Wang**, B. Zhou. The partial Legendre transform in Monge-Ampère equations. Preprint.

[6] **L. Wang**. Interior Hölder regularity of the linearized Monge-Ampère equation. *Calc. Var. Partial Differential Equations*, **64** (2025), no. 1, Paper No. 17.

[5] **L. Wang**, B. Zhou. $C^{1,\alpha}$ regularity of variational problems with a convexity constraint. Preprint.

[4] **L. Wang**, B. Zhou. Liouville theorems for a class of degenerate or singular Monge-Ampère equations. *J. Geom. Anal.*, **34** (2024), no. 11, Paper No. 352.

[3] Y. H. Kim, N. Q. Le, **L. Wang**, B. Zhou. Singular Abreu equations and linearized Monge-Ampère equations with drifts. To appear in *J. Eur. Math. Soc. (JEMS).*

[2] **L. Wang**, B. Zhou. Interior estimates for Monge-Ampère type fourth order equations. *Rev. Mat. Iberoam.*, **39** (2023), no. 5, 1895–1923.

[1] H. G. Li, J. -N. Wang, **L. Wang**. Refined stability estimates in electrical impedance tomography with multi-layer structure. *Inverse Probl. Imaging*, **16** (2022), no. 1, 229–249.

Honors and scholarships

2024 China National Scholarship

Presidential Scholarship (Peking University)

Outstanding Research Award (Peking University)

2023 Presidential Scholarship (Peking University)

Merit Student (Peking University)

2022 Exceptional Award for Academic Innovation (Peking University)

Talks

July 21-27, 2024 Workshop on Geometric Analysis 2024, School of Mathematical Sciences of Inner Mongolia University, Hohhot, China

March 24, 2024 Ph.D. Mathematics Forum, School of Mathematics and Statistics of Wuhan University, Wuhan, China

March 30, 2022 Geometric PDE seminar, Academy of Mathematics and Systems Science of the Chinese Academy of Sciences, Beijing, China

Teaching experience

Spring 2024 **Teaching assistant, Advanced math B (PKU)**

Fall 2023 **Teaching assistant, Mathematical analysis III (PKU)**

Spring 2023 **Teaching assistant, Mathematical analysis II (PKU)**

Fall 2022	**Teaching assistant, Mathematical analysis I (PKU)**
Spring 2022	**Teaching assistant, Arithmetic of elliptic curves (PKU)** **Teaching assistant, Diophantine approximation (PKU)**
Fall 2021	**Teaching assistant, Mathematical analysis I (PKU)**
Spring 2021	**Teaching assistant, Functional analysis (PKU)**
Fall 2020	**Teaching assistant, Advanced math C (PKU)**